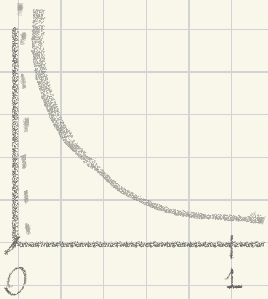
Sep 22 warmup

Find all values of $p>0$ for which $\int_0^1 \frac{1}{x^p}\,dx$ converges

(limit will have a value and won't be $\pm\infty$)

when $x=0$, function will be undefined - discont

$[0,1]$, discont at 0



$$\int_0^1 \frac{1}{x^p}\,dx = \lim_{t\to 0^+} \int_t^1 \frac{1}{x^p}\,dx$$

$$\frac{1}{-p+1}(x)^{-p+1}$$

$$\lim_{t\to 0^+}\left(\frac{1}{-p+1}(x)^{-p+1}\right)\Big|_t^1$$

$$= \lim_{t\to 0^+}\left(\left(\frac{1^{-p+1}}{-p+1}\right) - \left(\frac{t^{-p+1}}{-p+1}\right)\right)$$

If $p=1$ then the denominator cannot be 0 because it's discontinuous

Also if $p=1$, antiderivative ends up being $\ln|x|$ and since $\ln|0|$ is undefined there will always be ∞

If $p>1$, $-p+1<0$ and $\lim_{t\to 0^+} t^{-p+1} = \infty$

why?

So the only option is $p<1$

If $p<1$, $-p+1>0$ and $\lim_{t\to 0^+} t^{-p+1} = 0$